

Bedirhan Avdanci

PERSONAL INFORMATION

EMAIL: bedirhan.avd@gmail.com / bavdanci@uni-bremen.de
PERSONAL WEBSITE: <https://bedirhanavdanci.github.io/>

RESEARCH INTEREST

Microbial Ecology, Environmental Genomics, Computational Ecology

LANGUAGE

Turkish (Native) Azerbaijani (Native) English (8.0/9 IELTS) Persian (Elementary)

EDUCATION

- 2026 – PRESENT Master of Science (M.Sc.) in Marine Microbiology, **Max Planck Institute for Marine Microbiology (MarMic)**, Bremen, Germany
- 2022 – 2025 Bachelor of Science (B.Sc.) in Biology, **Hacettepe University**, Ankara, Turkey
CGPA: 3.94/4.00, valedictorian, highest ever in the whole faculty
Thesis: The effects of climate change and land-use on belowground communities over time.
Advisors: DR. MARIE SÜNNEMANN, PROF. NICO EISENHauer

PUBLICATIONS

3. **Avdanci, B.**, Edis, G., Deniz, S. A., Tüfekcioğlu, I., Yıldızcan E. E., Keskin, E., Tavsanoğlu, C. Fire Filters Soil Microbiome Assembly and Drives Microbial Functional Restructuring in a Mediterranean Pine Forest. (*Manuscript in preparation*)
2. Sünnemann, M., **Avdanci, B.**, Ferlian, O., Huang, Y., Lochner, A., Constantinou, I., Gajona, V., Meiners, A., Merk, J., Smolarek, E., Tremmel, R., Ziebart, E., Eisenhauer, N. Long-term Drought and Fertilization Reduce the Soil Microbial Respiration Response to a Simulated Heatwave. *Applied Soil Ecology*. (*Under Review*)
1. Usta, N., Tüfekcioğlu, I., **Avdanci, B.**, & Tavsanoğlu, C. (2026). The Effect of Elevation on Forest Structure and Tree Community Composition in Trojan Fir Forests in Kazdağları (NW Anatolia, Türkiye). *Applied Ecology and Environmental Research*, 24(1), 989-1002.

ORAL AND POSTER PRESENTATIONS

1. **Avdanci, B.**, Edis, G., Deniz, S. A., Yıldızcan, E. E., Keskin, E., Tavsanoğlu, C. Community Composition and Resistance Levels of Soil Bacteria and Fungi in a Mediterranean Forest Under Varying Wildfire Severities. *Oral presentation in English at: XI. Ecology and Evolutionary Biology Symposium, 8-11 July 2025, Ankara, Türkiye*

TEACHING

- 2026 Guest Lecturer for Introduction to R Programming in Natural Sciences Workshop, 1-2 June, Cankiri Karatekin University

AWARDS AND FELLOWSHIPS

2026 - 2028	M.Sc. Scholarship by Max Planck Society
All Semesters	High Honor Student at Hacettepe University

RESEARCH EXPERIENCE

2022 JULY – 2025 JUNE



Functional Ecology Lab, Hacettepe University Undergraduate Internship

Engaged in ecological investigations within steppe and Mediterranean ecosystems, conducting field-based soil and plant sampling alongside laboratory processing of ecological specimens. Organized, and analyzed preliminary datasets, applying ecological and statistical methods to extract meaningful patterns. Contributed to the preparation of research materials, supporting the development and refinement of ongoing ecological research.

ADVISOR: PROF. CAGATAY TAVSANOGLU

2025 FEBRUARY – MARCH



Evolutionary Genetics Lab, Ankara University Undergraduate Researcher

Performed DNA extractions, PCR amplifications, and library preparation for an eDNA metabarcoding investigation of post-wildfire soil microbial communities. Processed soil samples and executed laboratory workflows for high-throughput sequencing. Organized the dataset and independently analyzed bacterial and fungal community responses across gradients of wildfire severity.

HOST: PROF. EMRE KESKIN

2024 JULY – SEPTEMBER



Eisenhauer Lab, German Centre for Integrative Biodiversity Research (iDiv) Erasmus+ Traineeship Programme Internship

Contributed to long-term ecological experiments (DroughtNet) evaluating the effects of drought and mineral fertilization on soil bacterial and fungal communities. Conducted soil sampling, PLFA-based laboratory analyses, and soil processing for microbial community characterization. Organized and analyzed datasets and contributed to scientific writing within the Experimental Interaction Ecology group.

ADVISORS: DR. MARIE SÜNNEMANN, PROF. NICO EISENHAUER

2023 APRIL – NOVEMBER



TUBITAK (Scientific and Technological Research Council of Türkiye) Research Assistant (TÜBİTAK 1002-A Scholarship Holder)

Project: The Impact of Post-Fire Restoration Practices on Plant Communities in Turkish Pine (*Pinus brutia*) Forests of Mugla, Marmaris, Turkey

Contributed to fieldwork and data collection assessing post-fire restoration impacts on *Pinus brutia* forest communities. Assisted with vegetation sampling, ecological assessments, dataset preparation, and the analysis of plant community responses to restoration practices.

ADVISORS: PROF. CAGATAY TAVSANOGLU, DR. GOKHAN ERGAN

SKILLS

PROGRAMMING LANGUAGES:	R (Advanced), Python (Intermediate), Bash (Intermediate)
SCRIPTING AND TYPESETTING:	LATeX
METHODOLOGICAL EXPERTISE:	Targeted Amplicon Sequencing (DNA extraction, PCR amplification, gel electrophoresis, library preparation, high-throughput sequencing, and bioinformatic analysis) Phospholipid Fatty Acid Analysis / PLFA (lipid extraction, methylation, gas chromatography, and microbial community profiling)

MEMBERSHIP

2024 – PRESENT	Ecology and Evolutionary Biology Society of Türkiye (<i>Member of the Current Supervisory Board</i>)
2021 – 2022	Co-Founder of the Ankara University Biological Sciences Society

ACADEMIC COURSES, WORKSHOPS, AND CERTIFICATES

- 2025 Land Ethics Education Project, 10-11 May, Ankara, Türkiye
- 3rd introduction to R programming in Natural Sciences Workshop (Ecology and Evolutionary Biology Society of Türkiye)
- R Programming A-Z™: R For Data Science with Real Exercises! (UDEMY)
No: UC-d8cbdf48-79fa-41a8-94e0-81173-b35a201

SCIENCE COMMUNICATION & WRITING

- 2026 Aşağıdan, Yukarıdan Yolun Sonu Görünüyor! [Link](#)
- 2026 Çobanlık, Yağ tenekesi, Akademi ve Başvurular. [Link](#)
- 2025 Unutulmuş bir Tanrı: Benim Sadık Yârim Kara Topraktır. [Link](#)
- 2024 Diana Wall. [Link](#)
- 2024 Super-Small Predators in Soils: Who Are They and What Do They Do? [Link](#)
(Translation from English to Turkish; published in Frontiers for Young Minds).
- 2024 The Fascinating World of Belowground Communication [Link](#)
(Translation from English to Turkish; published in Frontiers for Young Minds).

SCIENTIFIC MEETINGS

- 2026 International Day for Biological Diversity, 22 May, Ankara, Türkiye
- 2026 Beyond COP31: Peoples' Climate Summit Meeting, 15 May, Ankara, Türkiye
- 2026 Climate Workshop for Youth: Towards COP31, 15 May, Ankara, Türkiye
- 2025 XI. Ecology and Evolutionary Biology Symposium, 8-11 July, Ankara, Türkiye
- 2025 International Wildlife Documentary Film Festival, 15-18 May, Ankara, Türkiye
- 2025 Birds and Words: Tracing biocultural diversity in sub-Saharan Africa, 27 April, Ankara, Türkiye
- 2024 IUCN red list from past to present and new approaches and trends in conservation biology, 4 November, Ankara, Türkiye
- 2024 Effects of global climate change on Salt Lake plants, and habitat diversity Conference, 5 March, Ankara, Türkiye
- 2024 Biodiversity of Anatolia, from Steppes to Mountains Conference, 14 February, Ankara, Türkiye
- 2023 IX. Ecology and Evolutionary Biology Symposium, 17-19 July, Istanbul, Türkiye
- 2022 VIII. Ecology and Evolutionary Biology Symposium, 7-9 September, Ankara, Türkiye
- 2022 III. Multidisciplinary neuroscience conference, Ankara, Türkiye
- 2022 Lewontin's Legacy, Science, Philosophy and Policy Conference, 9-10 April, Ankara, Türkiye
- 2021 5th Biogeography Conference, 16 April, Ankara, Türkiye
- 2021 16th Aykut Kence Evolution Conference, 22-23 February, Ankara, Türkiye

REFERENCES

Prof. Dr. Cagatay Tavsanoğlu
Hacettepe University, Department of Biology, Division of Ecology, Functional Ecology Laboratory
Email: ctavsan@hacettepe.edu.tr / ctavsanoğlu@gmail.com

Prof. Dr. Nico Eisenhauer
German Centre for Integrative Biodiversity Research (iDiv), Experimental Interaction Ecology Laboratory
Email: nico.eisenhauer@idiv.de

Dr. Marie Suennemann
German Centre for Integrative Biodiversity Research (iDiv), Experimental Interaction Ecology Laboratory
Email: marie.suennemann@idiv.de